LAW OF SINES AND LAW OF COSINES

PLANE TRIGONOMETRY

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

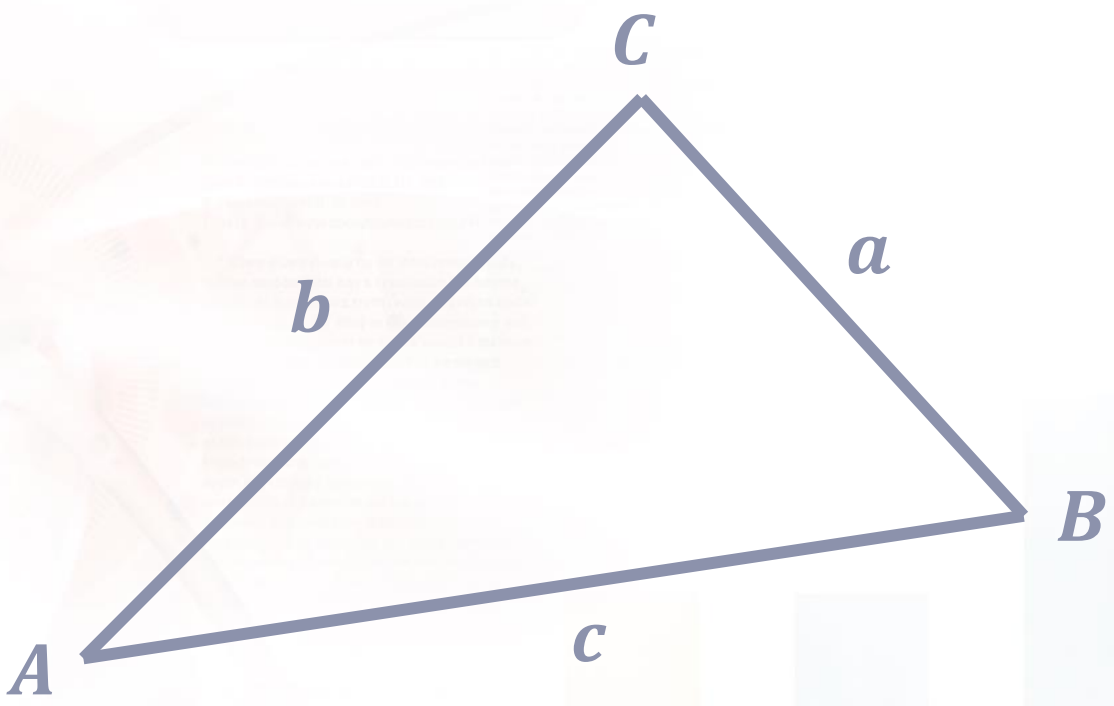
Law of Sines

Law of Cosines

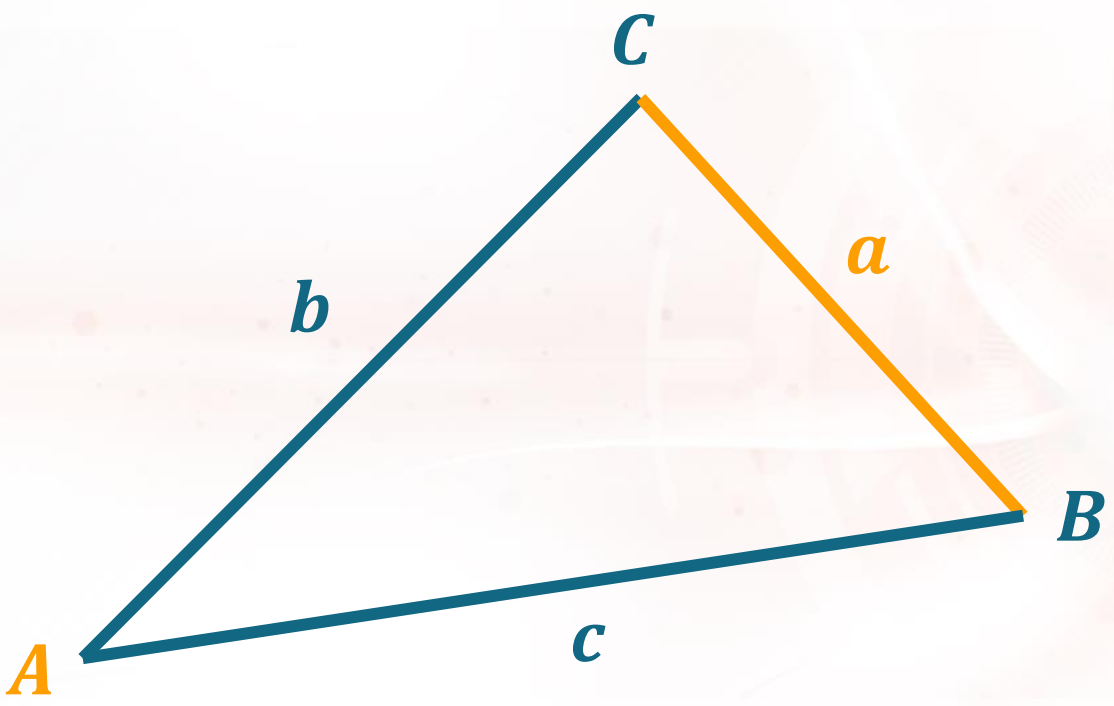
LAW OF SINES

OBLIQUE TRIANGLE

Triangle that has **no right** angles

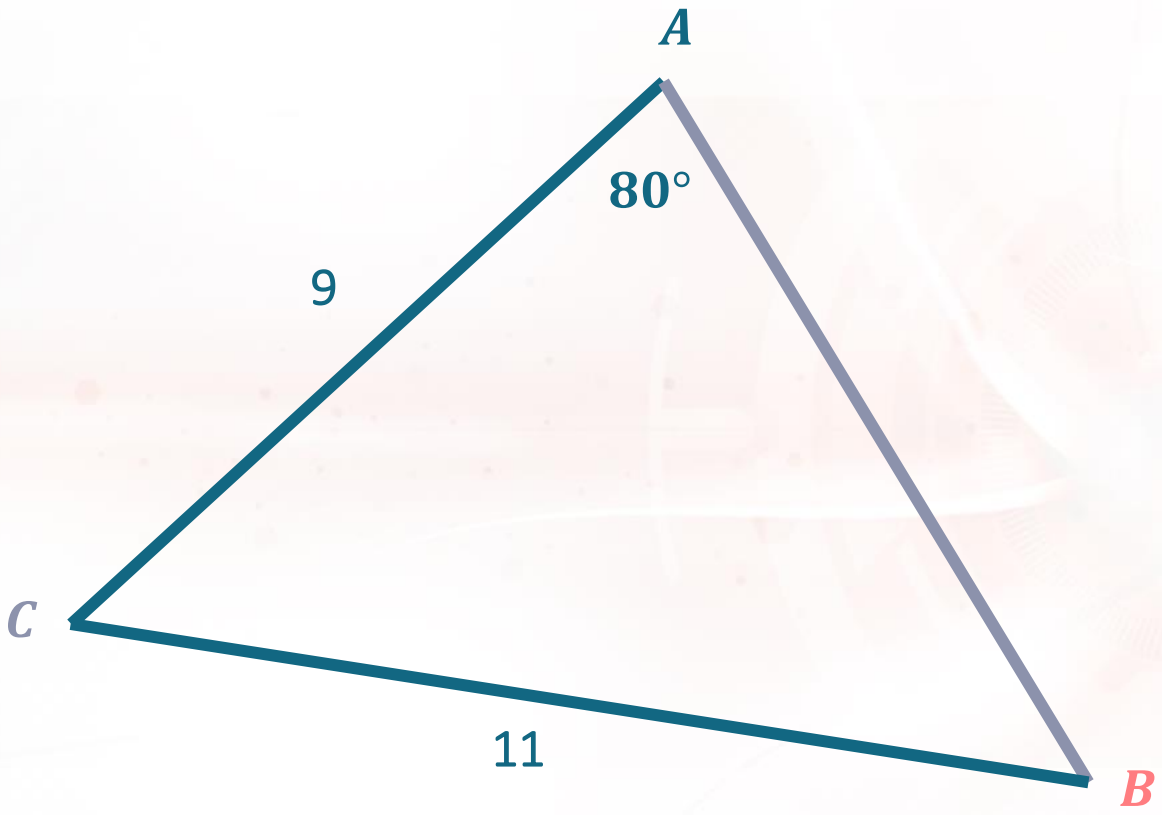


LAW OF SINES



$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

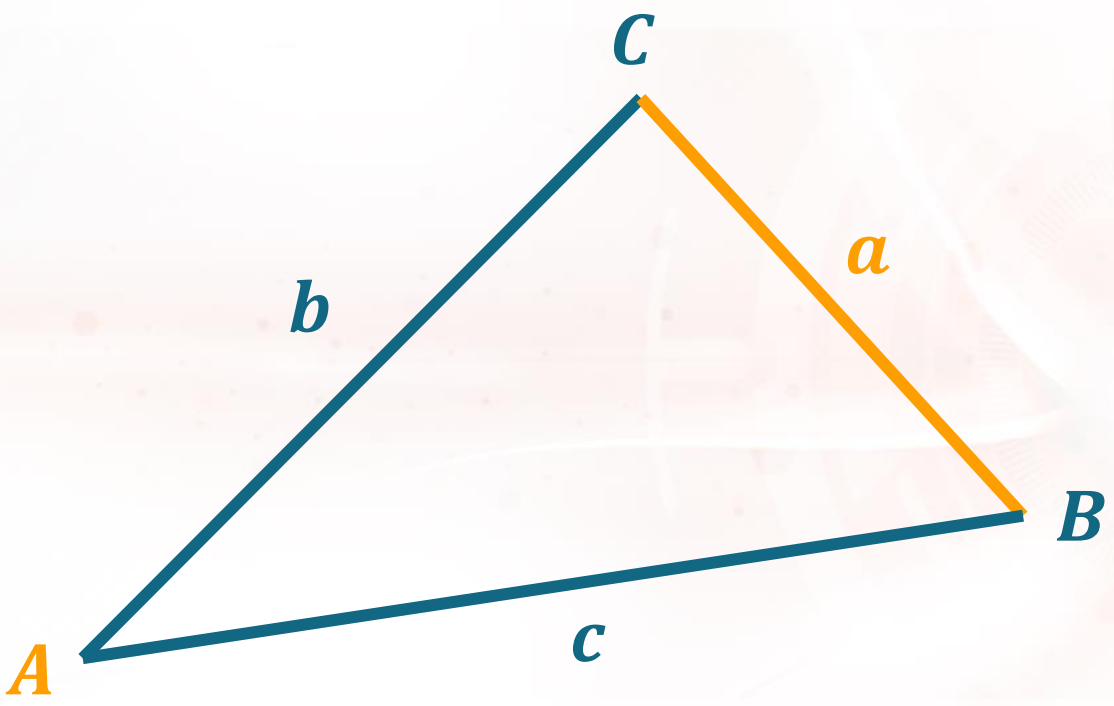
Find $m\angle B$.



A man starts his morning walk at a point A reaches two points B and C and finally back to A such that $\angle A = 60^\circ$ and $\angle B = 45^\circ$, $AC = 4\ km$ in the triangle **ABC**. Find the **total distance** he covered during his morning walk.

LAW OF COSINES

LAW OF COSINES

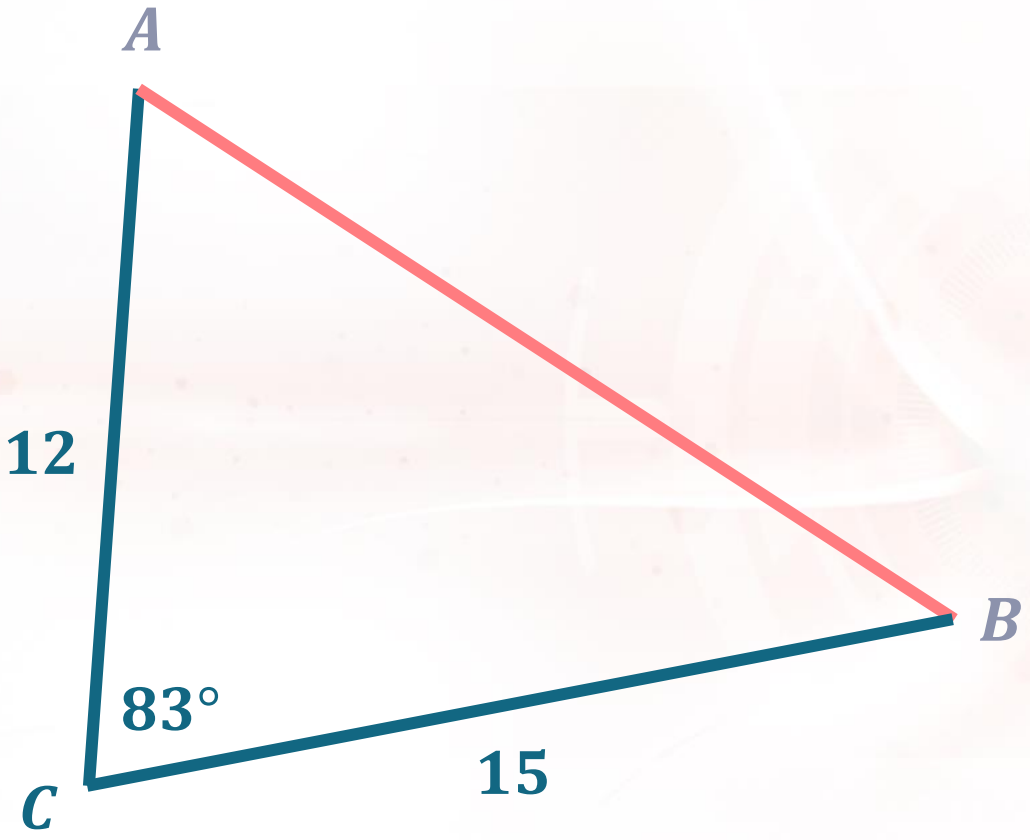


$$a^2 = b^2 + c^2 - 2bc\,cosA$$

$$b^2 = a^2 + c^2 - 2ac\,cosB$$

$$c^2 = a^2 + b^2 - 2ab\,cosC$$

Find AB.



EXERCISE

A plane is **1 *km*** from one landmark and **2 *km*** from another. From the planes point of view the land between them **subtends** an angle of **45°**. How far **apart** are the landmarks?

SEATWORK